

Lecture 30: Line integrals in space, curl, exactness and potentials

Line Integrals: Vector field $\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$; $d\vec{x} = \langle dx, dy, dz \rangle$
(force)

Curve C in space: Work = $\int_C \vec{F} \cdot d\vec{x} = \int_C Pdx + Qdy + Rdz$

Evaluate: parametrize C , express x, y, z in terms of a parameter

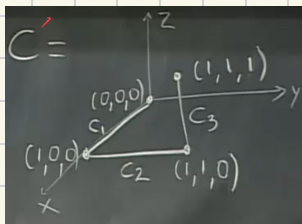
Example: $\vec{F} = \langle yz, xz, xy \rangle$

$C: x=t^3, y=t^2, z=t, 0 \leq t \leq 1$

$dx=3t^2dt, dy=2tdt, dz=dt$

$$\int_C \vec{F} \cdot d\vec{x} = \int_C yz dx + xz dy + xy dz = \int_0^1 t^3 \cdot 3t^2 dt + t^4 \cdot 2t dt + t^5 dt = \int_0^1 6t^5 dt = \left[t^6 \right]_0^1 = 1$$

Example 2: Same $\vec{F}$, C'



$$\begin{aligned} \int_{C'} &= \int_{C_1} + \int_{C_2} + \int_{C_3} \\ &= 0 + 0 + 1 = 1 \\ \int yz dx + xz dy + xy dz &= 0 \end{aligned}$$

For C_1, C_2 : are in the xy -plane, therefore $z=0, dz=0$

For $C_3: x=1, y=1$ From 0 to 1
 $dx=0, dy=0$

$$\int_{C_3} \vec{F} \cdot d\vec{x} = \int_0^1 xy dz = \int_0^1 dz = 1$$

In fact, $\vec{F}$ is conservative. $\int_C \vec{F} \cdot d\vec{x} = \int_{C'} \vec{F} \cdot d\vec{x}$
 $\vec{F} = \nabla(xyz) = \langle yz, xz, xy \rangle$

If we know this, the fundamental thm of calculus for line integrals:

$$\int_C \nabla f \cdot d\vec{x} = f(P_1) - f(P_0)$$

Here $f(1,1,1) - f(0,0,0) = 1 - 0 = 1$

Test for gradient fields:

$$\vec{F} = \langle P, Q, R \rangle \stackrel{?}{=} \langle f_x, f_y, f_z \rangle$$

$$\text{If so, then } \begin{cases} P_y = f_{xy} = f_{yx} = Q_x \\ P_z = f_{xz} = f_{zx} = R_x \\ Q_z = f_{yz} = f_{zy} = R_y \end{cases}$$

Criterion: $\vec{F} = \langle P, Q, R \rangle$ (defined in a simply connected region) is a gradient $\Leftrightarrow P_y = Q_x, P_z = R_x, Q_z = R_y$

Example: for which a and b is: $axy dx + (x^2 + z^3) dy + (byz^2 - 4z^3) dz$ exact?

$$\begin{aligned} P_y = ax &= 2x = Q_x \Rightarrow a=2 \\ P_z = 0 &= 0 = R_x \text{ AND} \\ Q_z = 3z^2 &= bz^2 = R_y \Rightarrow b=3 \end{aligned}$$

Find potential? Two methods: ① $f(x_1, y_1, z_1) = \int_C \vec{F} \cdot d\vec{x}$ (+constant) where C from $(0,0,0)$ to (x_1, y_1, z_1)
(same as in 2D)

② Antiderivatives: want: $f_x = 2xy, f_y = x^2 + z^2, f_z = 3yz^2 - 4z^3$

$$* f_x = 2xy \Rightarrow f = x^2 y + g(y, z)$$

$$* f_y = x^2 + z^2 = x^2 + g_y, \text{ so } g_y = z^2 \Rightarrow g = yz^2 + h(z)$$

$$* f_z = 3yz^2 - 4z^3 = yz^2 + h'(z) \text{ so } h'(z) = -4z^3, h = -z^4 + c$$

$$\text{So: } f = x^2 y + yz^2 - z^4 (+c)$$

$$\text{So for we had: } f = x^2 y + g = x^2 y + yz^2 + h(z)$$

Curl in 3D: If $\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$

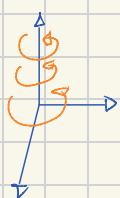
$$\text{curl } \vec{F} = (R_y - Q_z)\hat{i} + (P_z - R_x)\hat{j} + (Q_x - P_y)\hat{k}$$

If $\vec{F}$ is defined in a simply-connected region, $\vec{F}$ conservative $\Leftrightarrow \text{curl } \vec{F} = 0$

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \text{ We've seen: } \nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \quad \nabla \cdot \langle P, Q, R \rangle = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = \text{div } \vec{F}$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \hat{i} - \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \right) \hat{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \hat{k} = \text{curl } \vec{F}$$

Geometrically: "curl measures rotation component in a velocity field"



Rotation around z -axis at angular velocity ω

$$\vec{v} = \langle -\omega y, \omega x, 0 \rangle; \text{ curl } \vec{v} = 2\omega \hat{k}$$